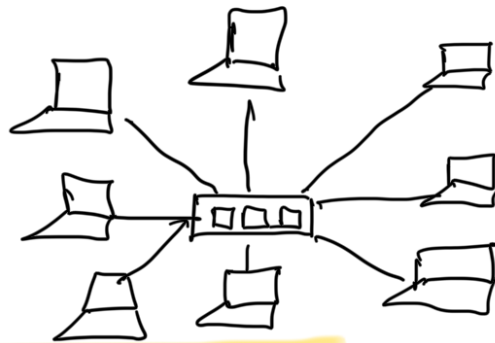


Introducing LAN technologies :-

Local Area Network topologies :-

Over the years, there have been experimentation and Implementation of various network designs. In reference of networking, when we refer to term "topology", we are actually referring to design or look of network at hand. Let's discuss Advantages and disadvantages of Topologies.

Star Topology :-



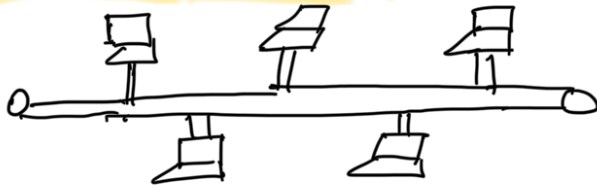
- * main premise of star topology is that devices are individual connected via central networking devices such as switch or router. This topology is most commonly found today because of its reliability and scalability despite of cost.
- * Any information sent to device in this topology sent via the central device to which it connects. Explore some Advantages and disadvantages of this topology below:
- * More cabling and purchase of dedicated networking equipment is required for this topology. More expensive than any other topologies.
- And Advantage is Topology is much more scalable in nature, that means more devices you can ^{add} as the demand for network increases.
- * Unfortunately, more network scale, more the maintenance is required

to keep network functional.

Ex:- If centralized hardware that connects devices fails, these devices will no longer be able to send or receive data. Thankfully the centralized hardware devices are often robust.

Bus topology: -

This type of connection relies upon a single connection which is known as backbone cable. This type of topology is similar to leaf off a tree in the sense that devices (leaves) stem from where branches on this cable.



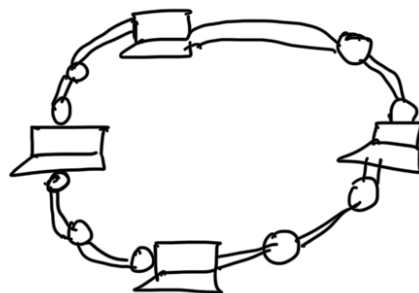
• Because all data destined for each device travels along same cable, it is very quickly prone to becoming slow and bottlenecked if devices within topology are simultaneously requesting data.

• It is difficult to identify which device is facing issue with data all travelling along same route.

• However bus topologies are more cost efficient and easier topologies to set up because of their expenses.

• Disadvantage is in bus topology, if main backbone cable breaks, the entire network fails because every device relies on that single path to communicate.

Ring Topology: -



This ring topology also known as token topology boasts some similarities. Devices such as computers are connected directly to each other then loop. Meaning less or little cabling

required and less dependence on dedicated hardware such as within star topology.

* A star topology works by sending data across hop until it reaches to the destined device, using other devices along hop to forward the data.

* In ring topology, data moves in one direction around circle. If one device or cable fails, the entire network goes down. It's easy to fix but slow, as data must pass through every computer until it reaches the destination.

What is a Switch?

Switch is central hub that connects many devices (like PC and printers) together. Unlike older "hubs" that blast data to everyone, a switch is smart and sends data only to specific device that needs it, making network much faster & less crowded.

* Switches are usually found in larger networks such as businesses, schools or similar sized networks, where there are many devices to connect to network. Switches can connect large number of devices having ports of 4, 8, 16, 24, 32 and 64 for devices to plug into.

* Switches are more efficient than lesser counterpart (hubs/repeaters). Switches keep track of what device is connected to which port.

* Both switches and router are connected one another. Creates redundancy, which means providing backup paths for your data. If one cable or device fails, the data simply takes a different route, preventing entire network from going offline.



What is a Router?

* It's a router job to connect network and pass data between them.

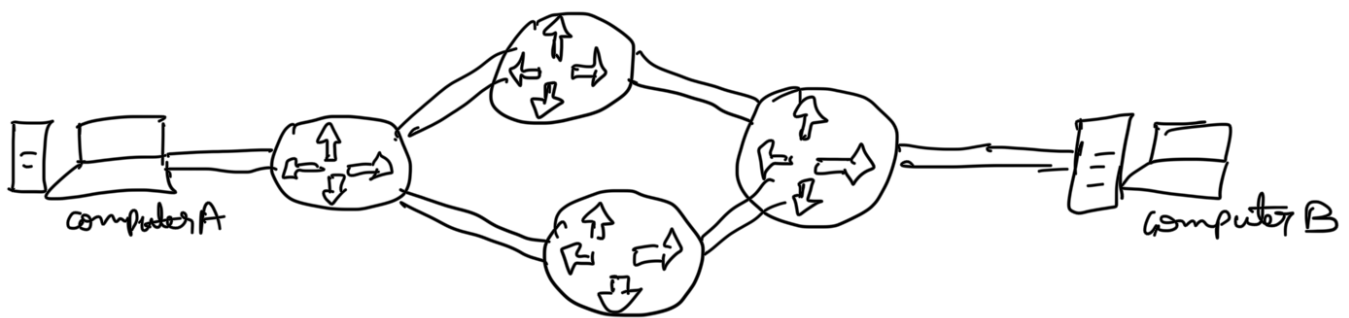
It does it by using routing.

* Router acts like traffic controller for Internet.

Connect different networks: It links to your home devices (like your phone or laptop) to much bigger network of Internet.

* When you send data, the router looks at all possible "roads" (paths) and picks the fastest, more reliable route to make sure your information reaches its destination.

* Routing is useful when devices are connected by many paths, such as in the example diagram below.



Practical task is done documenting.

Quiz:-

1) What does LAN stand for?

A) Local Area Network

2) What is the verb given to job that routers perform?

A) Routing

3) What device is used to centrally connect multiple devices on local network and transmit data to correct location?

A) Switch

4) What topology is cost efficient?

A) Bus topology

... topology

3) What topology is expensive to set up and maintain?

A) star topology

6) Complete the interactive lab attached to this task. What is the flag given at the end?

A) THE \$ TOPOLOGY - Fkws2